Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Database and ETL developer with 2+ years building **Python, SQL, Oracle, and Unix-based data systems**; experienced in batch and real-time processing, data modeling, cleansing, transformations, query tuning, QA, mappings, and production troubleshooting.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed and optimized **relational data models** across PostgreSQL and SQL Server, implementing schemas, indexed queries, transformations, and row-level isolation for 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built Python ETL workflows processing 5,000+ monthly jobs with incremental loads, cleansing, validation, reconciliation, error handling, and detailed failure states, achieving 99%+ production reliability through tested CI/CD releases.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Implemented **concurrent Python batch-processing workflows** for PDF, DOCX, image, and ZIP inputs, executing 1,000+ files per batch with validation, persistent results, fallback handling, and fault-tolerant processing.
- Performed QA across end-to-end UI, API, and database flows serving 150+ users and 10,000+ daily requests, reproducing complex interface failures and deploying tested fixes that reduced turnaround time 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Built **multi-source data ingestion and transformation pipelines** using Python, AWS Glue, S3, SQL, and caching, converting JSON, CSV, and Parquet inputs into normalized formats while improving processing efficiency 68%.
- Automated Unix/Linux-compatible batch builds, installations, tests, and reports through Jenkins and GitHub Actions, generating failure logs and coverage summaries across 20+ QA tests while reducing recurring manual entry 35%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Modeled and synchronized **2M+ structured records** across eight services and Azure SQL, defining schemas for users, resources, transactions, notifications, and reports while reducing cross-location data conflicts 33%.
- Traced end-to-end failures from React interfaces through Node.js APIs to Azure SQL, using Sentry, Azure Monitor, and Kibana to remediate issues and validate staged Docker releases through Jenkins.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Developed **batch and real-time data pipelines** over 50,000+ booking and payment events using Databricks, Spark, Flask, and Firebase, producing hourly analytics and maintaining sub-200-millisecond application responses.
- Documented end-to-end UI-to-database flows and 12+ technical data mappings for pricing, payments, status updates, and webhooks, implementing validation and idempotent reconciliation that reduced manual refund handling 70%.

Skills

Languages: SQL, Python, Bash, Java, JavaScript, TypeScript
Databases: Oracle, SQL Server, PostgreSQL, MySQL, Azure SQL, relational design, logical data modeling
Data Engineering: Batch ETL, real-time streams, data cleansing, transformations, mappings, validation, reconciliation
Performance & QA: Complex queries, indexing, query tuning, interface debugging, unit/integration testing, data quality
Delivery: Unix/Linux, shell environments, Git, GitHub Actions, Jenkins, CI/CD, technical specifications, AI-assisted workflows

Projects

Chat2SQL Code | *Oracle, SQL Server, Java, Docker*

- Enhanced an **Oracle and SQL Server database workspace** supporting 10+ systems with schema exploration, structure synchronization, query generation, data management, validation, Docker, Maven, and cross-platform troubleshooting workflows.

Big-Brain Live | *Python, PostgreSQL, Data Pipelines*

- Engineered a **data ingestion and indexing pipeline** processing 50,000+ records with incremental updates, metadata persistence, validation, REST APIs, and performance tuning, reducing retrieval latency 40% through optimized transformations.

FinNexus Live | *Next.js, Kubernetes, MongoDB*

- Built a **high-volume transaction data platform** supporting 50,000+ users and 1M+ transactions with secure third-party integrations, real-time updates, advanced filtering, MongoDB access, and Kubernetes-based CI/CD automation.